Problem Definition

Predicting a students (GPA) using a combination of academic and demographic factors of a student.

Objective: To build an end to end regression pipeline that models these relationships. This model helps ~~to~~ institutions to proactively identify ~~~~ students who need ~~support~~ support based on early semester results.

Dataset Discription

The project leverages a students performance tracking dataset.

Target variable → GPA

Features → Age, Gender, Study hours, attendance, major, part time Job.

Model Building

Implemented model → "Linear Regression"

This algorithm ~~is~~ is selected for its high interpretability. In a educational setting it allows administrators to read the exact directional weights (B coeffecients)

$$\boxed{\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_n x_n}$$

Model Evaluation

Metrics Tracked : (RMSE) Root Mean Squared Error and r^2 score (R^2) (Coefficient of Determination)

The RMSE explicitly identifies the standard deviation of the residuals, showing how close the ~~[illegible]~~ model's ~~prediction~~ predictions are to the actual unscaled GPA.

The R^2 score represents the proportion of variance in ~~[illegible]~~ students GPA that can be explained by input traits like attendance patterns.

Conclusion

The ~~[illegible]~~ model pipeline successfully handles structured string mapping, target features imbalance using scaling matrices. It applies inverse scale transformation.

Limitations → Linear models are unable to catch multi-variable ~~[illegible]~~ interactions naturally.

Future improvements → Introducing tree-based ensemble models (like Random forests)

Reflection :